

BUSS386 Problem Set 3

Forwards and Futures

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Problem 1 — Classification

For each contract below, state (a) forward or futures; (b) long or short for the named party; (c) exchange or OTC, with one short reason.

- (i) A Korean exporter agrees with its bank to sell USD 10M and receive KRW at ₩1,380 per USD, settling in six months.
- (ii) A retail trader buys five KOSPI 200 contracts on KRX at index level 405.
- (iii) A wheat farmer in Kansas locks in the October sale price of 50,000 bushels at \$5.80/bushel by trading at the Chicago Board of Trade.
- (iv) A jewellery maker enters a three-month contract directly with a Seoul bullion dealer to buy 1,000 oz of gold at \$2,300/oz.
- (v) An asset manager holding a \$50M S&P 500 ETF position takes a position in CME e-mini futures to neutralize equity risk.

Problem 2 — Margin accounts and daily settlement

You take a *long* position of one KOSPI 200 futures contract on Day 0 at index level $F_0 = 405.0$. Contract multiplier: ₩250,000 per index point. Initial margin: ₩10,000,000. Maintenance margin: ₩7,500,000. Daily settlement closes at the following index levels:

Day	0	1	2	3	4
Index close	405.0	408.0	402.0	393.0	397.0

- (a) Compute the daily mark-to-market gain/loss (in KRW) for Days 1–4.
- (b) Compute the end-of-day margin-account balance for each day, before any margin call.
- (c) Identify the day a margin call is triggered, and compute the deposit required to restore the balance to the initial margin.
- (d) Suppose you close the position on Day 4 at the closing index. Compute (i) the total cash you have deposited, (ii) the total cash returned to you when the position is closed, and (iii) your overall P&L on the trade.

Problem 3 — Long hedge with futures

Korean Air expects to buy 200,000 barrels of jet fuel on 15 May. On 1 March, the May jet-fuel futures price is \$90/bbl, and each contract is for 25,000 bbl. Assume the hedge is *perfect* (futures expiry coincides with the purchase date, and the underlying is exactly the same fuel).

- (a) What futures position should Korean Air take? How many contracts?
- (b) Compute the total cost on 15 May (in dollars) under each of these spot prices: $S_T \in \{\$84, \$90, \$96\}$ per barrel.
- (c) Explain why the total cost in (b) does not depend on S_T .
- (d) Name one realistic reason this hedge might *not* be perfect in practice.

Problem 4 — Short hedge, and the regret that follows

An Illinois corn farmer expects to harvest 250,000 bushels in December. On 1 July, the December CBOT corn futures price is \$4.60/bushel and each contract is for 5,000 bushels. Assume the hedge is perfect.

- (a) What futures position should the farmer take, and how many contracts?
- (b) Compute total December revenue for $S_T \in \{\$4.20, \$4.60, \$5.10\}$ per bushel.
- (c) The farmer's neighbour does not hedge at all. In which of the three scenarios does the neighbour end up better off, and by how much *per bushel*?
- (d) In December the price is \$5.10 and the farmer's futures position has lost money. A friend says the hedge was a mistake. In two or three sentences, say whether you agree and why.

Problem 5 — Basis risk on USD/KRW

On 1 March, Samsung Electronics signs a contract to receive USD 10,000,000 on *31 May* from a US customer. KRX lists USD/KRW futures expiring on the last business day of *March, June, September, and December*. Each contract is on USD 50,000.

Today (1 March):

- Spot USD/KRW $S_0 = 1,380$ (KRW per USD).
- June USD/KRW futures price $F_{\text{Mar}} = 1,390$.

- (a) What futures position does Samsung take, and how many contracts?
- (b) On 31 May, the spot is $S_{\text{May}} = 1,340$ and the June futures is $F_{\text{May}} = 1,370$. Compute (i) the futures P&L (in KRW), (ii) the gross KRW received from converting USD at spot, and (iii) the effective per-USD rate after the hedge.
- (c) Suppose, hypothetically, Samsung had received the USD on *30 June* (futures expiry) instead. What would the effective per-USD rate have been?
- (d) Explain in plain English what *basis risk* means in this trade and why the answers in (b) and (c) differ.

Problem 6 — Cross hedge / minimum-variance hedge ratio

A Korean airline expects to purchase 4,000,000 gallons of jet fuel in three months. Jet-fuel futures are not liquid, so it hedges with NYMEX heating-oil futures. From recent data:

- Standard deviation of monthly change in jet-fuel spot price: $\sigma_S = 0.025$ (\$/gallon).
- Standard deviation of monthly change in heating-oil futures price: $\sigma_F = 0.030$ (\$/gallon).
- Correlation $\rho = 0.85$.
- Heating-oil contract size: 42,000 gallons.

- Compute the minimum-variance hedge ratio h^* .
- How many heating-oil futures contracts should the airline trade, and on which side (long/short)?
- Compute the *hedge effectiveness* (the fraction of variance eliminated by the hedge).
- Why is the hedge effectiveness in (c) strictly less than 100%?
- Contracts trade only in whole numbers, so the position in (b) cannot match the target exactly. How many gallons of coverage does the rounded position actually provide, and how far is that from the minimum-variance target? Explain why the risk borne on that difference is *not* the same thing as the basis risk in (d).

Problem 7 — Hedging and risk measures

A Korean refiner will sell 100,000 barrels of gasoline in three months. At that date the gasoline price S_T is normally distributed with mean \$92/bbl and standard deviation \$8/bbl. Throughout, assume each futures price equals the expected future spot price, so that hedging is fair in expectation. Recall $\Phi^{-1}(0.05) = -1.645$.

- Unhedged.** Give the mean and standard deviation of revenue *per barrel*, and the 5% worst outcome (the revenue that is exceeded 95% of the time).
- Perfect hedge.** The refiner shorts gasoline futures expiring at T at $F_0 = \$92$. Give the mean, the standard deviation, and the 5% worst outcome of revenue per barrel.
- Cross hedge.** Gasoline futures turn out to be illiquid, so the refiner shorts *crude* futures instead, at $F_0^{\text{crude}} = \$78$. The crude price at T is normal with mean \$78 and standard deviation \$10, and its correlation with the gasoline price is $\rho = 0.75$.
 - Compute the minimum-variance hedge ratio h^* .
 - Compute the mean and standard deviation of hedged revenue per barrel.
Hint: revenue per barrel is $S_T - h^*(C_T - F_0^{\text{crude}})$, and $\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y$.
 - Compute the 5% worst outcome, and the hedge effectiveness.
- Put the three 5% worst outcomes side by side. In two sentences, describe what the refiner buys by hedging, and what it gives up by having to use crude rather than gasoline.

Problem 8 — Speculation and leverage

A trader is bullish on Korean equities. The KOSPI 200 index stands at **1,000**. She wants ₩1 billion of index exposure and is choosing between:

- (A) Buying ₩1 billion of a KOSPI 200 ETF outright, paid in cash.
- (B) Buying KOSPI 200 futures (multiplier ₩250,000 per index point). KRX requires initial margin of 10% of contract value.

Ignore dividends, fees and interest on the margin balance.

- (a) What is the notional value of *one* futures contract, and how many contracts give ₩1 billion of exposure?
- (b) How much cash must she post as initial margin under (B)?
- (c) The index rises 3%, to 1,030. For *both* (A) and (B), compute the profit in KRW and the percentage return *on the cash committed*.
- (d) Instead the index falls 3%, to 970. Repeat (c).
- (e) How far would the index have to fall — in index points and in percent — to wipe out the entire initial margin under (B)?
- (f) The trader's view and the index's behaviour are identical in (A) and (B). Where, then, does the amplified return come from, and what has she given up in exchange?